BAHRAIN POLYTECHNIC

بـولـيـتـكـنـك البحريـــن

Assessment Cover Sheet

Assessment Title	Unity Game: Project (Group) + Logbook / sheet		
Assessment Type	Controlled	Group	Not must-pass
Due Date	See table	**Course Code**	IT8101
Course Title	Games Development		
Internal Moderator's Name			
External Examiner's Name			

Instructions:

1. Requests for extensions should be made 2 working days before the deadline to the Course Coordinator. Extensions will only be approved with valid reasons.
2. Late submissions will be marked out of 60%.
3. You are only permitted a maximum of one extension per course per semester.

Learner ID		**Date Submitted**	
Learner Name			
Programme Code			
Programme Title			
Lecturer's Name			

By submitting this assessment for marking, I affirm that this assessment is my own work.

Learner Signature	

Do not write beyond this line. For assessor use only.

Assessor's Name			
Marking Date		**Maks Obtained**	

Comments:

Contents

1. Learning Outcomes

The following learning outcomes will be assessed in this group of assessments:

1. Demonstrate critical knowledge in computer graphics principles and game development concepts, and apply them in a specialized game environment (Game Project and SCRUM Logbook)
2. Work in teams to create games to a given brief following SCRUM methodology (Game Project and SCRUM Logbook)
3. Follow best practice, industry standards, professional ethics, using specialized programming and documentation skills and conventions during the programming process (Game Project)

2. Unity3D Game Project

Phase	Weightings	Due Date	Type	Submission
Part 1 – **Game Project**	35%	5/24/2026	Group	Electronic/Physical
Part 2 – **SCRUM Logbook**	15%	5/21/2026	Individual within Group	Electronic

***The weightings for the project components are taken from the course gradebook and form 50% of the total course grade**

3. Game Implementation

The task is to create and submit a game created using Unity3D. You will work in this project **in groups of 5.** The game story and genre will be decided by the development team, and its features will have to target:

1. The basic required features as set below, and
2. A set of custom features that will depend on the game genre and will be agreed upon between the development team and the tutor.
3. SCRUM progress

3.1. The basic requirements (features)

1. The game must have **an introductory screen** that includes **a Main Menu**, allowing the user to:
    a. Start a new game
    b. View instructions
    c. View credits
    d. Settings to configure the game (Audio, Resolution…etc.), and
    e. Quit the game
2. The game must include a minimum of **playable levels equal to the team size**.
    a. If the game focuses on advanced and complex mechanics, the number of levels can be negotiated with the tutor, but this must take place when submitting a game design document.
    b. If the game is Open-World (i.e. an RPG), you can work on different biomes within the same level which in turn would be considered as levels, but each one of them needs to be distinct.
3. The game must include **at least one animated character**, either controlled by the player or AI.
    a. The character must have multiple animation states that are triggered either using input or as a scripted behavior.
4. The game must use one of the **custom camera** types (third person, first person or observation camera)
5. The game must allow appropriate **user input based on the selected export platform**.
6. The game must have a **reward system** (score, points, coins, etc.) In case of making a game that doesn't rely on rewards, you can add features that resemble it. (i.e Speedrun System)
7. The game should include **sound effects, music, environment sound** where appropriate.
8. The game should include some **special effects** (e.g. particle effects, post-processing, etc.)
9. The game should be **exported** and tested on a selected platform. (PC, Web)
10. The game should be **bugs-free** and **playable in terms of completeness** (i.e. players can beat the game / demo)

3.2. The custom requirements (features)

These must be features not included in the set above and will be selected by the development group based on the game genre, idea, and story. The group is expected to draft a list of prospective custom features and discuss them with the tutor. The final list of the custom features will be agreed upon among all members of the group and the tutor to fulfil the marking requirements as set below. You must have 1 advanced custom feature and 5 additional custom features. An example of an advanced feature could be Open World Streaming, Systemic Gameplay, LLM Integration, Multiplayer...etc. An example of a simple custom feature would be symbolic AI, Inventory System...etc.

You must <u>**include all requirements**</u> (basic and custom) in your *product backlog*, following SCRUM methodology **(see below 3.3 and Appendix: SCRUM).**

3.3. SCRUM Methodology

You are expected to apply SCRUM methodology during your implementation phase. You must have weekly plans to accomplish a set of tasks that are selected from the product backlog, which holds all the project's requirements. We use SCRUM to simulate a real game development studio's workflow. The proper implementation of SCRUM can accelerate your work, allowing you to spot issues earlier, and accomplish tasks faster. **(Read More: Appendix)**

3.4. Best Practice

1. You are expected to apply SCRUM methodology during the implementation phase.
2. You are expected to have a well-organized project folder. All game assets should be kept within their appropriate sub-folders (e.g. scripts saved in the “Scripts” folder, scenes saved in the “Scenes” folder, etc.)
3. Your code should look clean and be well documented.

3.5. Marking Scheme

The following table lists the high-level breakdown for the Game Project and Logbook requirements. A more detailed analysis can be seen in the published marking scheme.

Section	Mark
Basic Features	40
Custom Features	30
SCRUM Practice and Logbook	30
Total	100

4. Deliverables

Scrum

Product Backlog and Game Design Document

Before you start developing your game, you are supposed to submit a product backlog that includes all features/requirements (basic and custom). Your weekly sprints will be based on the requirements in this product backlog.

You are also expected to submit a small game design document describing your game. Please follow the guidelines included in the game design document template provided through Moodle.

Weekly Sprint Logs

During the project implementation phase, you are expected to submit a sprint log every week. Each sprint log submission must contain:

Sprint Log

The sprint log shows the set of features or requirements that your team will be working on <u>the following week</u>. It also shows the group members responsible for each task, and the estimated time you expect to take to complete each task. **(See Appendix)**

Game Submission

The group needs to submit a zip file to Moodle containing:

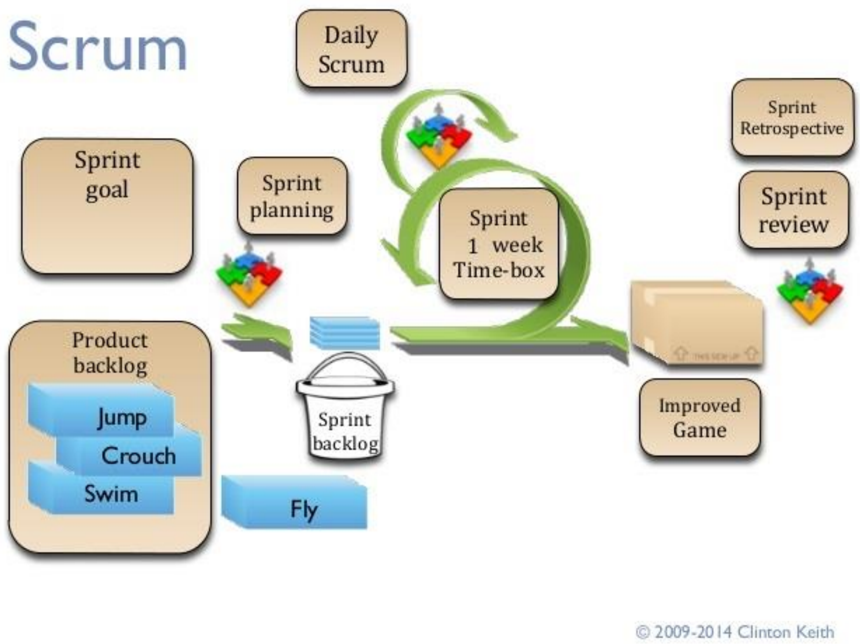
a. Unity project folder (containing all assets used), and
b. The final build of the game (example: game.exe)

Moodle submission is a must. Contact your tutor if your project's size is too large for a Moodle submission. In this case a cloud drive submission may be allowed.

5. Appendix

Scrum

Scrum is a goal-oriented team configuration that enables iterative and rapid development. The way Scrum works allows for constructive feedback to be received and tackled very early in the development phase, ensuring an improved result. Ideally, a Scrum team would be made of a product owner, a scrum master, and the developers.



Your team is expected to prepare a product backlog (list of requirements) sorted by priority (high to low). You will then select a number of requirements from your product backlog to complete in a Sprint, which will be tracked in a Sprint backlog (backlogs templates available in appendix). Each team member must take responsibility for any number of Sprint backlog items of his choice. Backlog items should be split equally between team members.

You are expected to have weekly deliverables in the form of Scrum Sprints. A Sprint is a collaborative work from all team members to achieve a certain goal. Sprints should focus on certain requirements. Your whole project would be split into many Sprints that break down your project into smaller, much clearer goals. **(see Appendix 2 and 3)**

After each Sprint, you are expected to present your results to the product owner (your tutors) in a Sprint Review. The product owner will evaluate and either accept or reject the presented results.

Product Backlog

Feature ID	Feature	Description
CH001	*Character Control*	*Character should be able to move 360 degrees based on user input.*

Sprint Backlog

Sprint 1: Sprint Title (e.g Game Design, Character Control, etc…)				
Feature ID	**Feature**	**Member Responsible**	**Est Time**	**End of Sprint Result** ***(For tutors to fill)***
CH001	*Character Control*	*Ahmed Ali*	*2 Days*	